

Generation Interconnection System Impact Study Report

For Queue Project Q12-0188

Lasalle 100 kV
December 01, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0188. The Interconnection Customer proposes a 152.7 MW wind facility interconnecting at the Lasalle 100 kV substation in the Illinois area. The request was received on December 21, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 121 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,386,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lasalle – Dahlia 100kV
Contingency: Lasalle – Centro Medico 6008 100kV L
Loading: 118.4% (Rating: 131 MVA)
DFAX: 100.0%
Pre-project loading: 1.4%
2. Facility: Wayne Trace – Darlington Rd 100kV line
Contingency: Dealb – Richland 500kV
Loading: 118.0% (Rating: 111 MVA)
DFAX: 6.3%
Pre-project loading: 109.4%
3. Facility: Lasalle – Centro Medico 6008 100kV
Contingency: Lasalle – Dahlia Substation 100kV
Loading: 117.6% (Rating: 131 MVA)
DFAX: 100.0%
Pre-project loading: 1.4%
4. Facility: Segs 5 – Hillcrest 100kV
Contingency: Richland – Dekalb 500kV
Loading: 108.0% (Rating: 181 MVA)
DFAX: 7.5%
Pre-project loading: 101.7%
5. Facility: Slocum Rd – Dekalb 230kV L
Contingency: Richland Substation – Dekalb 500kV
Loading: 107.6% (Rating: 508 MVA)
DFAX: 26.9%
Pre-project loading: 99.5%
6. Facility: James De Young – St. Lawrence Substation 230kV Line
Contingency: E Methuen – Lonetree 500kV
Loading: 105.3% (Rating: 578 MVA)

DFAX: 7.4%

Pre-project loading: 103.4%

7. Facility: Allegheny Energy Units 3 4 & 5 500/1 kV XFMR

Contingency: Hilltop – Piney Crk Project Substation 500kV L

Loading: 105.2% (Rating: 821 MVA)

DFAX: 10.8%

Pre-project loading: 103.2%

8. Facility: Norwich – Larson Street 230kV

Contingency: Barrett – Norwich 500kV

Loading: 103.7% (Rating: 738 MVA)

DFAX: 9.7%

Pre-project loading: 101.7%

9. Facility: E Methuen 500/230 kV Autotransformer

Contingency: Swan Valley – Trego 500kV

Loading: 102.9% (Rating: 981 MVA)

DFAX: 13.9%

Pre-project loading: 100.7%

10. Facility: Pickford – Adams 230kV Line

Contingency: Trego – Beaver 500kV

Loading: 100.1% (Rating: 531 MVA)

DFAX: 8.8%

Pre-project loading: 97.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lasalle – Dalia 100kV	Rebuild line with higher-capacity conductor	\$441,000	100% (\$441,000)
Wayne Trace – Darlington Rd 100kV line	Rebuild line with higher-capacity conductor	\$3,637,000	2% (\$81,000)
Lasalle – Centro Medico 6008 100kV	Rebuild line with higher-capacity conductor	\$1,150,000	100% (\$1,150,000)
Segs 5 – Hillcrest 100kV	Reconductor line	\$997,000	2% (\$22,000)
Slocum Rd – Dekalb 230kV L	Reconductor line	\$22,115,000	2% (\$501,000)
James De Young – St. Lawrence Substation 230kV Line	Reconductor line	\$38,490,000	2% (\$857,000)
Allegheny Energy Units 3 4 & 5 500/1 kV XFMR	Replace transformer with higher-rated unit	\$29,655,000	2% (\$694,000)
Norwich – Larson Street 230kV	Reconductor line	\$5,867,000	2% (\$134,000)
E Methuen 500/230 kV Autotransformer	Replace transformer with higher-rated unit	\$15,895,000	2% (\$354,000)
Adams – Pickford 230kV	Reconductor line	\$6,658,000	2% (\$152,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$4,386,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.